

Interview Planning

Advice: Try not to be nervous

I will divide my background into:

- Education
- Experience
- Research

Education

- Bachelor's in Electronic Engineering
- MSc in Machine Learning
- PhD in Data Science & Statistical Modeling

Industrial Experience

2017 – Intelligent Electronic Solutions

- C++ / Object-Oriented developer
- Development of gambling machines

2020–2021 – Machine Learning Engineer at SUPSI-DTI (Switzerland)

- Developed deep learning pipelines for object detection systems
- Developed software modules for data acquisition and preprocessing

2021–2024

While working on my PhD, I also collaborated with xFarm as a consultant (20–30%)

- Implemented and fine-tuned object detection systems and computer vision algorithms (YOLO, segmentation)

Main challenges:

- Lack of data
- Generation of synthetic images
- Data augmentation

2024– until now

- Designed modular software components for integrating AI vision models into production systems
- Developed multi-agent and agent-based orchestration frameworks for agronomic applications
- **Applications included:**
 - Weather information
 - Insect pest control
 - Irrigation systems

Technologies used:

- LangGraph
- LangChain

Later, we decided to continue working with Google ADK (a framework for agent/tool-use APIs).

We experimented with:

- Embeddings
- Fine-tuning
- Guardrails
- Evaluation metrics like LLM-as-evaluator
- Trajectory Success

Challenges handled:

- Hallucinations
- Language switching for users
- Unexpected agent behavior

Research

2018–2020 – Machine Learning Research in Soundscape Ecology
I used machine learning to automatically detect changes in natural soundscapes by finding hidden patterns in audio recordings.

Work:

- Developed an Automatic Acoustic Heterogeneity Identification methodology using unsupervised GMM
- Identified acoustic clusters to discover latent behaviors and associate them with functional animal/landscape characteristics

Challenge Identified:

- determining the optimal number of clusters was difficult

2021–2024

Based on this research gap, I began working on an metric to determine the correct number of clusters.

The approach modeled uncertainty by analyzing the probability of belonging to a class together with geometric distances among distributions.

I worked on a joint project between:

- University of Antioquia, La Rochelle University, France

This resulted in:

- **Rendon et al. (2023)** – *Uncertainty Clustering Internal Validity Assessment using Fréchet Distance*

During this work, we discovered that ecosystem sound patterns could be used as a language to understand ecosystems themselves.

This led to:

- **Rendon et al. (2025)** – *Letting Ecosystems Speak for Themselves: An*

Unsupervised Methodology for Mapping Landscape Acoustic Heterogeneity

Environmental Modelling & Software

In this work, I used clusters to find correlations between acoustic patterns and:

- Taxonomy of species
- Geographical locations
- Satellite imagery

2024–Until now

This gave me the idea that patterns of the ecosystems could be understood as a language

And with the Colombian university we already submitted

We also created a knowledge base for theoretical concepts in Landscape Ecology: Terminology, Fragmentation, and Reductionism of terms in this area of ecology

We are currently in review process of an article on this work:

- **Graph NN for acoustic patterns**
- (which can be understood as a knowledge graph, to understand the relationship y sounds and ecosystem transformation /species

Personal Interest

Doing that developed a strong interest in language-related AI topics in general, including:

- Structure and ontologies
- LLMs
- RAG systems

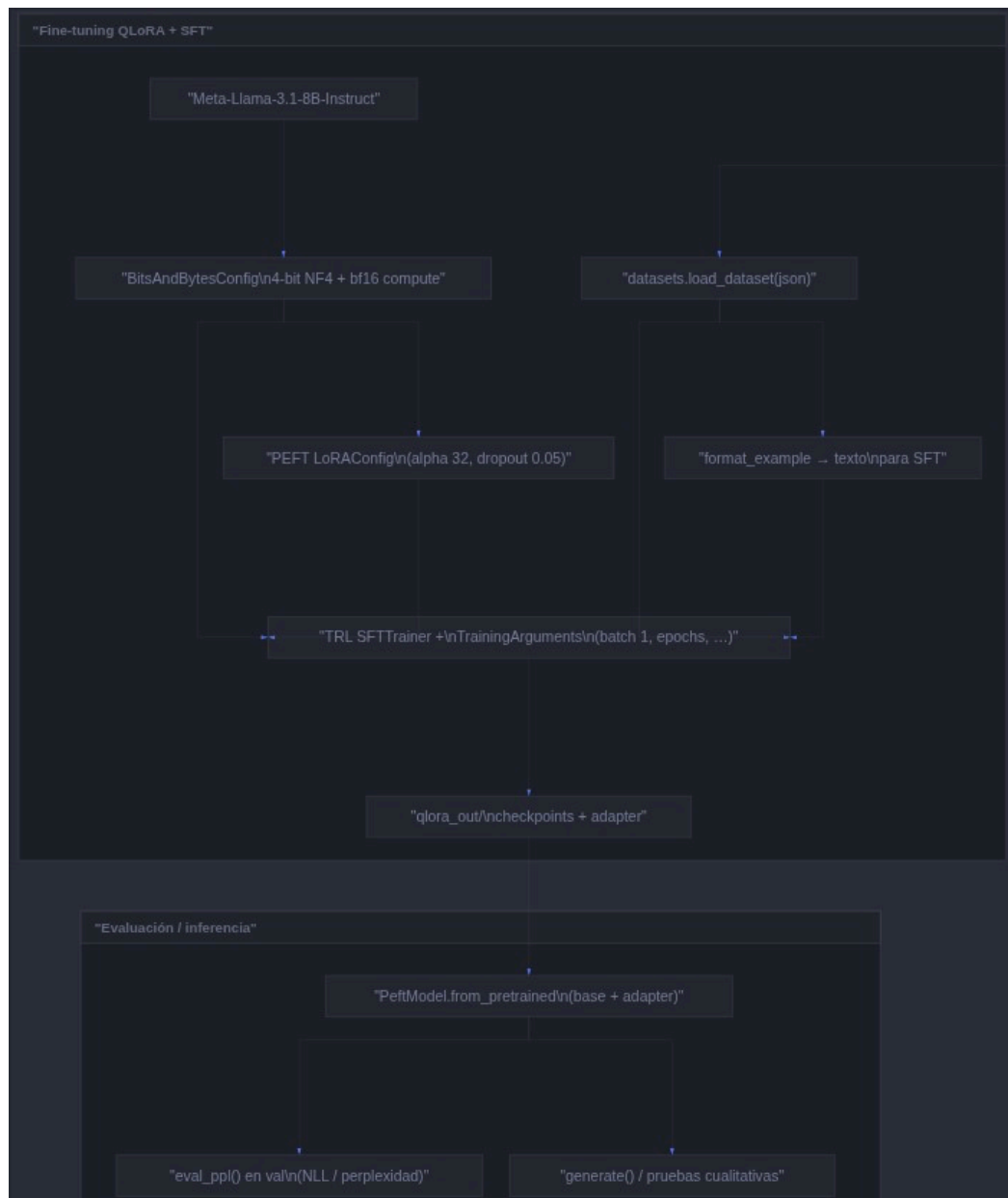
In my free time, I enjoy experimenting with:

- Fine-tuning models
- Creating my own agents
- Using Ollama and Hugging Face
- Exploring local LLM deployment and adaptation

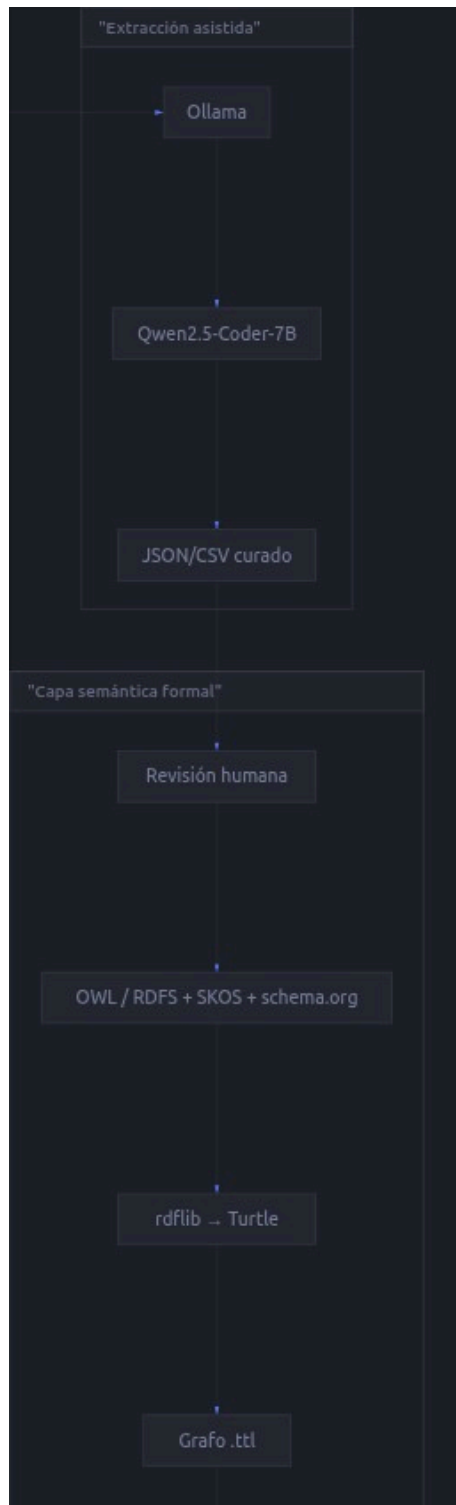
From predictions/embeddings you might infer:

- (Location_A, has_activity_level, High)
- (Location_B, acoustically_similar_to, Location_C)
- (Location_D, belongs_to_cluster, Cluster_2)

These could be stored as graph triples.







Core Relationship

Knowledge Graph = Structured Data Representation

A knowledge graph stores facts as a graph:

- **Nodes** = entities/concepts
- **Edges** = typed relations between entities

Example:

(Paris) —[capital_of]→ (France)

Knowledge graphs are typically symbolic and human-interpretable.

Graph Neural Network = Learning Model for Graph Data

A GNN is a neural architecture designed to learn from graph-structured inputs by passing messages between connected nodes.

It produces embeddings/representations for:

- nodes
- edges
- entire graphs

GNNs are one of the main deep learning tools for graph data

Embeddings/predictions are:

- numerical latent representations
- learned statistics/patterns
- not structured semantic facts

A knowledge graph requires **explicit symbolic relations**.

When Could the Output Become a Knowledge Graph?

Only if you **convert model outputs into explicit relational knowledge**.

For example:

Example 1: Derived Semantic Relations

From predictions/embeddings you might infer:

- (Location_A, has_activity_level, High)
- (Location_B, acoustically_similar_to, Location_C)
- (Location_D, belongs_to_cluster, Cluster_2)

These could be stored as graph triples.

Example 2: Learned Ecological Relationships

If your model discovers patterns like:

- dawn-active sites cluster together
- certain habitats share acoustic signatures

You can formalize them into relations.

Important Distinction

GNN Output:

Learned latent representations / predictions

Knowledge Graph:

Structured symbolic relational database of facts

The GNN can **help build** a knowledge graph, but does not inherently produce one.

